A Bit-Exact Fixed-Point FPGA Echo State Network for Lithium-Ion Battery SOC Estimation with an Exploratory SOH Readout

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***Abstract*—Model-based state-of-charge (SOC) observers need an equivalent-circuit model, an open-circuit-voltage curve, and the present cell capacity, while data-driven recurrent networks are costly on embedded hardware. We study whether an echo state network (ESN) with a closed-form linear readout can estimate competitively without battery-model identification while remaining cheap in fixed-point hardware. A 100-neuron Q6.10 reservoir with a SOC readout and an exploratory state-of-health (SOH) readout was implemented on an Artix-7 field-programmable gate array (FPGA) and verified bit-exact against an independent golden model in simulation and on the board. On the NASA Prognostics Center of Excellence (PCoE) dataset, the estimator is evaluated on 2,201 discharge windows across 29 cells, with each evaluation cell held out in turn. In an offline, window-level evaluation without any capacity input, the ESN reaches a SOC mean absolute error (MAE) of 0.067 and a root-mean-square error (RMSE) of 0.086 and shows no statistically significant difference in per-cell MAE from an extended Kalman filter (EKF) given the rated capacity (MAE 0.074, $p = 0.51$) or a capacity-tracking EKF (0.061, $p = 0.31$). The deployed reservoir exhibits 1.7–2.8× lower inter-cell standard deviation of per-cell MAE than the evaluated EKF configurations ($p \leq 0.007$). An EKF given each cell's measured capacity is more accurate (0.063), and on four reference cells a capacity-tracking EKF performs best. The SOH readout is evaluated separately on reference and heterogeneous cell populations and is weaker on the latter. The reservoir core uses 318 lookup tables (LUTs) and two digital signal processing (DSP) slices; causal online estimation is not yet demonstrated.**

***Index Terms*—Battery management system, echo state network, extended Kalman filter, fixed-point arithmetic, FPGA, lithium-ion battery, reservoir computing, state of charge, state of health.**

I. Introduction

State of charge (SOC) and state of health (SOH) cannot be read from a sensor [1], so every battery management system (BMS) infers them from voltage and current, and each classical route carries an assumption that ageing eventually breaks. Coulomb counting needs the present capacity, and extended Kalman filter (EKF) observers [2], [3] need an equivalent-circuit model whose parameters change with temperature and capacity fade. Deep recurrent networks such as long short-term memory (LSTM) [4] avoid explicit models but need floating-point arithmetic, backpropagation, and memory that a small embedded BMS does not have. An echo state network (ESN) [5], a form of reservoir computing [6]–[8], trains only a linear readout in closed form, so its weights can be quantized once and stored on chip. This paper asks whether an ESN can estimate battery state competitively without battery-model identification and still map efficiently onto fixed-point field-programmable gate array (FPGA) hardware. ESNs have been implemented on FPGAs before (Section II), so the contribution of this paper is the combination of six elements:

1) a non-spiking, fixed-point ESN (Q6.10 reservoir) with closed-form ridge readouts in Q4.14, for SOC, with an exploratory SOH readout;
2) a resource-efficient Artix-7 implementation with one time-multiplexed multiply-accumulate (MAC) core, block RAM (BRAM) weights, a double-buffered state memory, and a lookup-table tanh, using 318 lookup tables (LUTs) and two digital signal processing (DSP) slices;
3) bit-exact verification against an independent golden model in simulation and on the programmed board;
4) a cross-cell evaluation holding out each of 29 NASA Prognostics Center of Excellence (PCoE) cells (4–44 °C) in turn;
5) a comparison with EKFs given the rated or measured capacity and with a capacity-tracking EKF;
6) an analysis of cross-cell error consistency with the cell as the statistical unit.

II. Related Work

Plett's EKF [2] and its sigma-point variant [9] are the reference model-based estimators; both also describe joint state and parameter estimation, which can track capacity online, and Li *et al.* [3] separate SOC and SOH onto two EKF timescales. Yun *et al.* [10] implement a fixed-point EKF for SOC on an FPGA; like any model-based observer it relies on identified cell parameters and a capacity value. LSTMs [4] and transformers [11] remove the explicit model but are evaluated in floating point; quantized recurrent accelerators [12] target backpropagation-trained networks. Reservoir computing is reviewed in [8], [13]. The closest battery work, Kamarudin *et al.* [14], estimated SOC with a spiking leaky integrate-and-fire reservoir on the NASA PCoE cells B0005–B0007 and B0018, simulated in software (BindsNET), with a multilayer-perceptron readout trained by stochastic gradient descent, a feedforward network as the only baseline, and energy inferred from spike counts rather than measured. Digital reservoirs, ESNs included, have already been built on FPGAs [15], [16]. To the best of our knowledge, existing studies have not demonstrated this combination of bit-exact fixed-point FPGA ESN implementation, board-level verification, cross-cell battery evaluation, and comparison against capacity-aware EKF

configurations (Section I).

III. ESN Formulation and Fixed-Point Design

The reservoir state $\mathbf{x}(t) \in \mathrm{R}^N$ evolves as

$$x(t+1) = tanh\left(W_{in}u(t) + Wx(t) + b\right) \quad (1)$$

where $\mathbf{u}(t) \in \mathrm{R}^4$ holds measured voltage, current, temperature, and load current. The hardware uses $N = 100$ and no leaky integration. $\mathbf{W}$ has 10% density with an 80%/20% excitatory/inhibitory split; after Q6.10 quantization it holds 996 nonzeros in $[-0.295, 0.295]$ with spectral radius 0.900, $\mathbf{W}_{\text{in}}$ lies in $[-0.495, 0.490]$, and the deployed $\mathbf{b} = \mathbf{0}$. SOC is read out per step and SOH once per window from the window-mean state $\bar{\mathbf{x}}$, optionally projected onto k principal components:

$$\hat{SOC}(t) = w_{SOC}^{\top}x(t) + c_{SOC} \quad (2)$$

$$\hat{SOH} = \beta^{\top}P_k(\bar{x} - \mu) + c_{SOH} = w_{SOH}^{\top}\bar{x} + c'_{SOH} \quad (3)$$

Because the projection is linear, (3) collapses to one 100-coefficient dot product. Coefficients are fitted by ridge regression on the states of the bit-exact model and stored as Q4.14 words, so the readouts are trained on exactly what the FPGA computes.

All datapath values are 16-bit signed Q6.10 (range $[-32, 32)$, resolution 2^{-10}). Q12.20 products are summed in a guarded wide accumulator and, after the 104 MACs of one neuron, rescaled with round-half-up,

$$s = \left(\Sigma + 2^9\right) \gg 10, \quad (4)$$

then saturated, biased, and clipped to $[-5, 5]$. Because tanh is odd, 5121 entries cover $[0, 5.0]$ and the sign is restored after lookup; the table's maximum error against double-precision tanh is 4.88×10^{-4}, half of one least-significant bit.

IV. FPGA Architecture and Verification

Fig. 1 shows the controller handshake that sequences each neuron update: the last recurrent MACs, the done strobe, and and the start of the next input-weight phase. Fig. 2 shows the top-level architecture, in which one compute core serves all 100 neurons. The address generator produces the input-weight (a_{in}), recurrent-weight (a_W), input ($a_u = tN_{\text{in}} + k$), and state-read (j) addresses, and two multiplexers feed one MAC core with ($\mathbf{W}_{\text{in}}$, $\mathbf{u}$) and then ($\mathbf{W}$, $\mathbf{x}$). For neuron i, the core performs four input and 100 recurrent MACs, rescales by (4), adds the bias, clips, looks up tanh, and writes $x_i(t+1)$.



Fig. 1. Controller handshake at the end of Run A: last recurrent MACs, done strobe, and next input-weight phase.

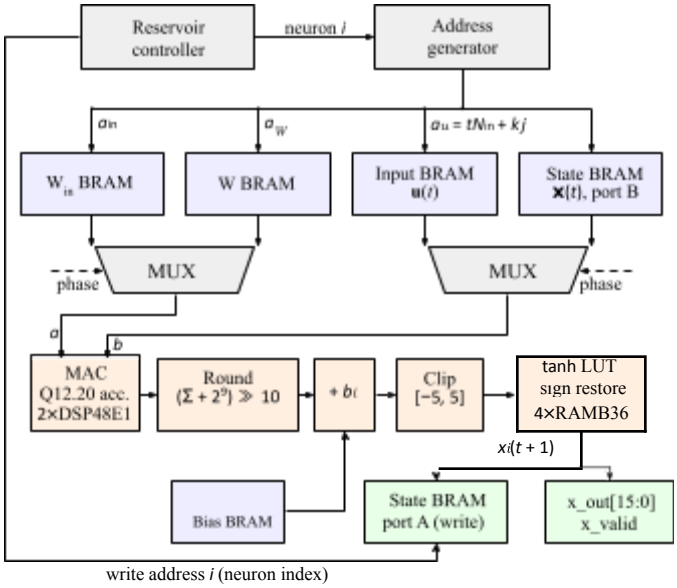


Fig. 2. Top-level FPGA architecture. The address generator produces the input-weight (a_{in}), recurrent-weight (a_W), input (a_u), and state-read (j) addresses; the controller's phase signal switches the MAC operands (a, b) from ($\mathbf{W}_{\text{in}}$, $\mathbf{u}$) to ($\mathbf{W}$, $\mathbf{x}$). Ports A and B belong to one dual-port state memory whose two buffers swap roles after each update.

The neuron core holds the MAC (two DSP48E1 slices) and the tanh read-only memory (ROM); six BRAM cores hold $\mathbf{W}$, $\mathbf{W}_{\text{in}}$, the inputs, the biases, and the two state buffers. Two indices are kept distinct: the neuron index i is the write address, while the recurrent read index $j \in \{0, \ldots, 99\}$ sweeps the previous state; merging them would silently remove the recurrent coupling. Because every neuron needs the complete $\mathbf{x}(t)$, the state memory is a simple dual-port block RAM split into two ping-pong buffers: port A writes one while port B reads the other, and the roles swap after each update. All memories total about 175 kbit, dominated by the densely stored recurrent matrix (160 kbit).

The Python golden model reimplements the register-transfer level (RTL) arithmetic exactly (weights, Q12.20 accumulation, rounding, saturation, tanh table) and was written from the specification, independently of the RTL. For every neuron update, both the golden model and the Vivado behavioral simulation log five values (rounded MAC, bias, biased sum, clipped tanh input, and tanh output), which are joined on (timestep, neuron) and compared word for word, localizing any discrepancy to a stage. *Run A* drives the deployed reservoir with a complete 64-step real discharge record (cell B0018, cycle 115; $64 \times 100 = 6400$ updates); *Run B* uses an earlier weight set with nonzero biases over two updates (200 updates) to exercise the bias path.

For board-level tests the core is wrapped with the two Q4.14 readouts and placed beside fixed-point EKF and unscented Kalman filter (UKF) cores on the same device, behind a universal asynchronous receiver-transmitter (UART) host interface. The host loads a *model pack* (readout weights, input scaling, and EKF/UKF tables) at run time, then streams each discharge window and receives SOC from all three estimators, the ESN SOH, and per-sample cycle counts. A software twin of the platform checks every window.

V. Experimental Protocol

A. Data and Estimation Window

The NASA PCoE dataset [17] contains 34 18650 cells and 2,794 discharge records at ambient temperatures of 4, 22-24, and 43–44 °C and discharge currents of 1, 2, and 4 A. The estimation window is the active discharge. Many records continue after the load is switched off; in this *post-load rest data* the current is zero, the voltage relaxes, and the reference SOC is constant, in some records for up to 98% of the samples. This tail lies outside the intended window, so each record ends at its last sample whose current magnitude exceeds 0.25 A. Internal pauses of pulsed loads (cells B0025–B0028) are kept. The rule is applied to every record, for training and testing alike, before any model is fitted (Section VI-C).

Windows were then rejected by fixed quality rules: fewer than 16 samples under load (143), capacity missing or below 0.2 Ah (88), or recorded capacity more than 10% from the integrated discharge charge (349). This leaves 2,214 valid windows. Each window was resampled to 64 points, min–max normalized with training statistics only, quantized to Q6.10, and fed to the reservoir, reset at each window start.

SOC labels come from trapezoidal Coulomb counting against the window's own measured capacity, so every label starts at exactly 1 and ends near 0: each window is assumed to begin fully charged, as the NASA protocol charges each cell to 4.2 V before a discharge; SOH labels are each window's capacity divided by the median of the same cell's first three valid capacities. This reference capacity is used only to construct labels, in the same way for training and held-out cells; it is not an input to the ESN, whose SOH readout sees only reservoir states. For a held-out cell these come from its earliest windows, so they precede the windows they normalize; scoring without those three windows per cell leaves the SOH mean absolute error (MAE) unchanged (0.046 on the reference cells, 0.065 on the cohort).

Six cells (B0033, B0036, B0040, B0049–B0051) are excluded from SOH training and scoring because their capacity records do not form a usable fade trajectory: four start with a partial first discharge (0.64–1.00 Ah, about half of the next), which corrupts the reference capacity; B0040 rises from 0.79 to 1.73 Ah; B0050 drops to 0.30 Ah and recovers; and three log 2.33–2.44 Ah, above the rated 2.0 Ah. They were flagged by a label-only screen (any SOH label outside 0.5-1.15), a plausibility check rather than a physical limit; in B0036 and B0051 a single window triggers it. SOH is therefore evaluated on 24 of the 29 cohort cells (1,746 windows; Section VI-G gives the sensitivity).

B. Cohorts

The *evaluation cohort* is the 29 cells with at least 10 valid windows (2,201 windows); B0041, B0049, and B0052 are used for training only, and B0045 and B0053 have no valid window. The *reference cells* B0005, B0006, B0007, and B0018 (24 °C, 2 A) are part of the evaluation cohort and also serve as the EKF calibration cells. In the cross-cell evaluation each cell is held out in turn; the ESN readout is trained on the 31 other cells with valid windows, and the EKF is calibrated on the reference cells (the other three when one is under test), so no held-out cell contributes to its own estimate. For the reference-cell evaluation, the ESN is trained only on the other three reference cells. In every fold an inner cross-validation over training *cells* selects the ridge parameter $\lambda \in \{10^{-4}, \ldots, 10\}$ and the principal component analysis (PCA) dimension by the root-mean-square error (RMSE) of the quantized Q4.14 readout. All estimators use identical windows, SOC labels, and preprocessing.

C. Baselines.

A one resistor-capacitor (RC) Thévenin EKF and UKF, in the same fixed-point form as on the board, use a fifth-order polynomial pseudo open-circuit voltage (OCV) and a least-squares R_0 (0.9–1.0 mΩ), with R_1 = 10 mΩ and C_1 = 3000 F fixed because the windows contain no relaxation. They run with either the cell's measured first-cycle capacity (an oracle value) or the rated 2.0 Ah. Calibrating them on all other cells degraded them (MAE 0.101), so the reference-cell calibration is used. A capacity-tracking joint EKF with state [SOC, V_1, Q] [2] starts each cell at the rated 2.0 Ah, models Q as a random walk, and carries it across windows in chronological order, in floating point. Coulomb counting with the measured capacity is also reported. All baselines start each window at the label's initial value, 1.0; the ESN receives no initial SOC (its state is reset to zero) and learns the full-charge start only from its training labels. Initializing the filters instead from the first sample's voltage through the fitted OCV and R_0 gives 1.0 in median (minimum 0.88) and changes no MAE by more than 0.001 or any paired conclusion. An initial error of 0.2 raises the EKF MAEs to 0.098–0.114.

D. Statistics.

Pooled metrics concatenate all 140,864 held-out steps. For comparisons the cell is the statistical unit: per-cell differences are given as a mean with a bootstrap 95% confidence interval (CI) and tested with the Wilcoxon signed-rank test [18], and the inter-cell spread of per-cell MAE is compared by a bootstrap ratio of standard deviations (SDs) and the Pitman–Morgan test for paired variances.

VI. Results

A. Bit-Exact Verification in Simulation

Both runs match at all five stages: 6400/6400 updates in Run A and 200/200 in Run B, with no missing, extra, or undefined records. Fig. 3(b) shows the comparison output: all 6400 neuron updates match at the MAC, bias, sum, tanh-input, and tanh-output stages, and all 64 SOC outputs and the SOH word of the readouts also match the golden model bit for bit.

B. Bit-Exact Operation on the Board

The platform was programmed on a Digilent Arty A7-100T (Fig. 3(a)) at 80 MHz and driven from a PC over UART. For cell B0018 (eight discharge windows over its life), ESN, EKF, and UKF SOC and the ESN SOH matched the software twin bit for bit in all eight windows, as did a run on the 4 °C cell B0047. Measured on chip, one sample costs 42 905 cycles for the ESN (536.3 μs), 245 for the EKF, and 662 for the UKF. The accuracies below are therefore the board's outputs.

C. Effect of Post-Load Rest Data

Table I recomputes the 29-cell metrics with and without post-load rest data. The with-rest set has 46 fewer windows (2,155 against 2,201) because, with the rest tail kept, the integrated current includes post-load samples and those 46 windows fail the capacity-consistency rule; on the 2,155 common windows, discharge-only MAE is 0.067 (ESN) and 0.063 (EKF). With the rest tail included, the ESN's largest error was 1.672, on B0050 cycle 10: after load-off, the zero current, recovering voltage, and a 0.75 V sensor dropout resembled no training window. Ending each window at the last loaded sample

lowers the ESN's maximum error to 0.636 and its 95th-percentile error from 0.219 to 0.168, and also improves the EKF.

TABLE I
SENSITIVITY OF THE 29-CELL SOC ERROR TO DATA PREPARATION

	ESN			EKF, meas. cap.		
Evaluation set (windows)	MAE	P95	Max.	MAE	P95	Max.
With post-load rest (2,155)[a]	0.090	0.219	1.672	0.073	0.266	0.778
Discharge-only, main (2,201)	0.067	0.168	0.636	0.063	0.209	0.478
excl. start <4.0 V (2,193)	0.066	0.166	0.549	0.063	0.210	0.478
B0056 c0 relabelled[b] (2,201)	0.067	0.167	0.599	0.063	0.210	0.478

[a]46 windows fail the capacity check if post-load current is integrated. [b]Capacity-based start 0.584, filters initialized there; sensitivity only. P95: 95th-percentile absolute error. EKF: reference-cell calibration.

The remaining maximum, 0.636 on B0056 cycle 0, is a labelling error. That first discharge starts at 3.84 V from a partially charged cell and delivers 0.785 Ah against 1.344 Ah in the next cycle, yet its label starts at 1 because labels are normalized to each window's own capacity. The ESN reads the low voltage; the filters follow the label because they start at 1.0. Table I therefore reports (1) the main evaluation with all windows and original labels; (2) the same without the nine windows that start below 4.0 V (eight in the test set); and (3) a separate sensitivity run in which only B0056 cycle 0 is relabelled from capacity to start at 0.584 (0.785/1.344 Ah) and end at 0, with every baseline initialized at 0.584. In (3) the ESN's MAE on that window falls from 0.289 to 0.104, the rated-capacity EKF's rises from 0.077 to 0.099, and the pooled ESN maximum becomes 0.599 (another low-start window). In (2) and (3) pooled MAEs move by at most 0.001 and no paired conclusion changes; all other results use the original labels.

D. SOC Accuracy and Cross-Cell Consistency (29 Cells)

Table II summarizes the 29-cell evaluation. The capacity-tracking EKF has the lowest pooled MAE (0.061), and the EKF given each cell's measured capacity follows (0.063); the ESN, which receives no capacity value, reaches 0.067, ahead of the EKF with the rated capacity (0.074). Per cell, the ESN shows no statistically significant MAE difference from the rated-capacity EKF (better on 18 of 29 cells; mean MAE difference +0.004, 95% CI [−0.012, 0.019], $p = 0.51$) or from the capacity-tracking EKF (12 of 29; −0.005 [−0.025, 0.016], $p = 0.31$), and it is less accurate than the EKF with the oracle capacity (8 of 29; $p = 0.011$). The same holds for per-cell RMSE and 95th-percentile error ($p \geq 0.34$). The fixed-point UKF, with the same calibration, is practically indistinguishable from the EKF (Table II): per-cell MAE differs by at most 0.002, and the paired tests against the ESN give $p = 0.55$ (rated) and 0.011 (measured capacity). A non-significant difference does not establish equivalence; no equivalence test was performed.

TABLE II
SOC ACCURACY ON THE 29-CELL COHORT (140,864 HELD-OUT STEPS)

	Pooled over steps					Per-cell MAE	
Estimator	MAE	RMS E	R^2	P95	Max.	SD	Worst
ESN (no cap.)	0.067	0.086	0.919	0.168	0.636	0.020	0.129
EKF, rated cap.	0.074	0.096	0.898	0.197	0.427	0.035	0.127
Cap.-tracking EKF	0.061	0.099	0.891	0.227	0.910	0.056	0.222
EKF, meas. cap.[a]	0.063	0.095	0.900	0.209	0.478	0.044	0.196
Coulomb count[a]	0.069	0.103	0.882	0.215	0.829	0.046	0.196
UKF, rated cap.	0.074	0.096	0.898	0.196	0.427	0.035	0.127
UKF, meas. cap.[a]	0.063	0.095	0.900	0.210	0.480	0.044	0.196

[a]Given each cell's measured capacity (oracle).

What distinguishes the ESN is its consistency across cells. Its per-cell MAE ranges from 0.043 to 0.129 (median 0.068, SD 0.020), whereas the EKF variants span 0.012–0.222 with SDs of 0.035–0.056. The capacity-tracking and oracle-capacity EKFs are better on typical cells (median 0.042 and 0.045) but fail more severely on others (worst cells B0028 at 0.222 and B0040 at 0.196); the rated-capacity EKF has a similar worst cell (0.127) but a higher median (0.074). The inter-cell SD ratio is 1.70 (95% CI [1.24, 2.67]) against the rated-capacity EKF and 2.75 ([1.70, 4.46]) against the capacity-tracking EKF; the Pitman–Morgan test rejects equal spread against every baseline ($p \leq 0.007$). The ESN also has the lowest 95th-percentile error (0.168 against 0.196–0.227), though its largest single-step errors exceed those of the rated-capacity EKF. The ESN is therefore competitive on average and more uniform across cells, but it is not more accurate.

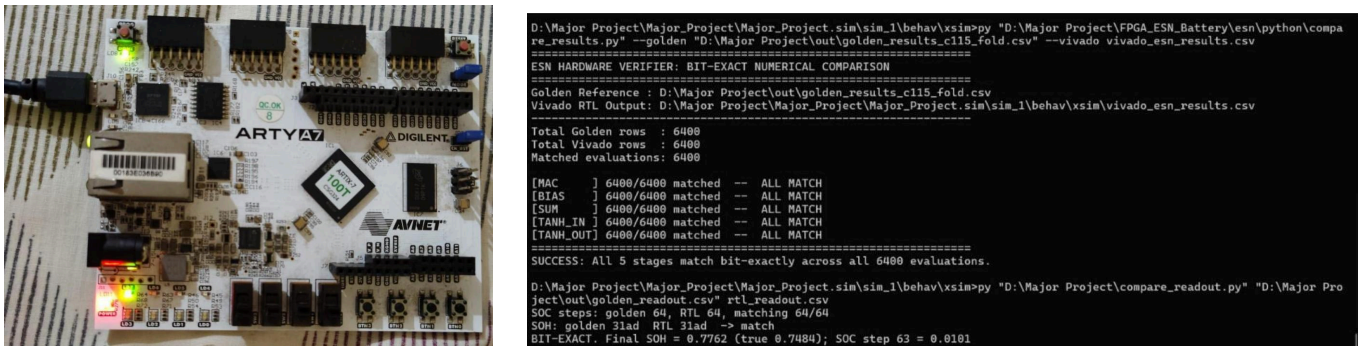


Fig. 3. (a) Arty A7-100T board running the ESN design. (b) Output of the comparison scripts for Run A: all five stages match over 6400 updates, and the 64 SOC outputs and the SOH word match the golden model.

E. Comparative Evaluation on the Reference Cells

With the ESN trained only on the other three reference cells, the capacity-tracking EKF performs best (MAE 0.051), followed by the EKF with the measured capacity (0.068, $R^2 = 0.917$) and the ESN (0.081, $R^2 = 0.894$), which beats the EKF only on B0005. From the rated 2.0 Ah, the joint EKF's capacity converges to within 0.08 Ah of the measured end-of-life value. Trained on 31 cells instead, the ESN gives 0.069 on these cells, level with the EKF (0.068), whose calibration is the same in both cases. On these four cells a ridge readout on the raw inputs does as well (R^2 0.896).

F. Training Coverage, Ablation, and Seed Variability

Training coverage matters. Trained only on the four 24 °C reference cells, the ESN does not transfer: SOC MAE is 0.633

on the other-load windows, 1.197 at 43–44 °C, and 0.233 at 4 °C, against 0.067, 0.075, and 0.059 when the readout is trained on all conditions, while the reference-calibrated EKF stays at 0.038–0.076. Adapting to new conditions changes only the model pack, not the reservoir: on the 4 °C cell B0047 the same bitstream gives 0.201 SOC MAE with the 24 °C pack and 0.043 with the all-condition pack, both reproduced exactly on the board.

On the 29 cells, a ridge readout on the four raw inputs reaches only MAE 0.110 (R^2 = 0.771), so across the heterogeneous cohort it is the reservoir that makes a linear readout work. A floating-point copy of the deployed reservoir reaches 0.063 (R^2 = 0.925); quantizing the reservoir to Q6.10 costs 0.002 MAE and the Q4.14 readout another 0.002. The inter-cell SD is 0.028 for raw inputs and 0.020–0.021 with any reservoir, and the cohort SOH R^2 rises from 0.077 to 0.20–0.22. Ten further reservoirs drawn from the same recipe gave pooled SOC MAE 0.073 ± 0.009 (0.066–0.091; R^2 = 0.899 ± 0.029). Their inter-cell SD stayed below the rated-capacity EKF's 0.035 for eight of ten draws but reached 0.054 for two, and the largest single-step error ranged from 0.63 to 1.13, so the deployed reservoir is among the better draws.

G. Exploratory SOH Readout

With each reference cell held out in turn, the SOH readout reaches a pooled MAE of 0.046, RMSE 0.059, and R^2 = 0.707, cutting RMSE by 40–62% per cell relative to predicting the training-set mean (per-cell R^2 0.44 for B0006 and 0.73–0.85 for the others); the joint EKF's capacity state, read as SOH, gives 0.116 MAE. Across the 24 SOH-evaluated cohort cells the ESN SOH is weaker: MAE 0.066 against 0.081 for the training mean, with R^2 = 0.217. Keeping the six excluded cells gives MAE 0.149 against 0.128 for the training mean and raises the MAE on the other 24 cells to 0.103 (mean 0.090); dropping only the 199 out-of-range windows gives 0.075 against 0.083. We therefore treat the SOH readout as exploratory.

H. Model Dependency and Hardware Cost

Table III summarizes the trade-off: the ESN needs no circuit model, OCV curve, or capacity value, but it needs labelled training windows from similar conditions and costs 65–175 times more cycles per sample on the chip.

TABLE III
MODEL DEPENDENCY AND HARDWARE COST OF THE COMPARED ESTIMATORS

	ESN	EKF / UKF	Cap.-tracking EKF
Circuit model, OCV curve	not needed	needed	needed
Capacity input	none	meas. or rated	rated, tracked
Training/calibration	31 cells	4 ref. cells	4 ref. cells
SOH output	readout	none	capacity state
On this FPGA	yes	yes	no (software)
Cycles per sample	42 905	245 / 662	n/a
Dynamic power[a] (W)	0.051	0.036 / 0.052	n/a

[a]Per block in the board platform, vectorless; ESN includes readouts.

I. Synthesis, Power, and Timing

Vivado 2023.1 post-route results for the reservoir core on an XC7A100T, with its outputs brought to ports so that synthesis keeps the full datapath, are in Table IV; the neuron core uses 259 of the LUTs and both DSPs. Timing closes at 80 MHz with 0.023 ns setup slack; the critical path is the single-cycle MAC. One update takes 42 807 cycles, because the schedule is fully serial with a single MAC: the reservoir core processes one resampled input point in 535.1 μs, below a nominal 10 ms BMS sampling period, and a 64-point window in 34.2 ms. The evaluation itself remains offline and window-based; real-time causal operation on streaming measurements has not been demonstrated.

TABLE IV
POST-ROUTE RESOURCES AND PER-UPDATE TIMING/POWER (ARTIX-7 XC7A100T)

Design, 80 MHz	Total power (W)	Dynamic power (W)	WNS (ns)	Latency (µs)	Energy[a] (µJ)
Reservoir core	0.120	0.023	+0.023	535.1	64.2
Board platform[b]	0.368	0.269	+1.317	536.3	n/a

Core: 318 LUTs (0.50%), 311 flip-flops, 11.5 BRAM (8.5%), 2 DSP48E1.
[a]Total power × latency. [b]ESN, readouts, EKF, UKF, UART, mixed-mode clock manager (MMCM; 0.122 W of the total); latency measured on chip. WNS: worst negative slack.

VII. LIMITATIONS

(i) The evaluation is offline: resampling each window to 64 points over its own length encodes its duration, every window starts from full charge by label construction, and the SOH feature needs the complete discharge; the hardware results establish implementation, bit-exact operation, and per-sample latency, which indicate potential for embedded BMS use; a real-time online estimator has not been demonstrated; and (ii) power is Vivado's vectorless estimate, not a measurement. Pipelining, parallel lanes, or skipping the zeros of **W** would cut the ESN's cycle count.

VIII. CONCLUSION

A 100-neuron Q6.10 ESN reservoir for SOC estimation, with an exploratory SOH readout, was implemented on an Artix-7 FPGA (318 LUTs, two DSP slices, 80 MHz) and verified bit-exact in simulation and on the board. On 29 held-out NASA PCoE cells, evaluated offline, it reached SOC MAE 0.067 without a capacity input or battery model, with no statistically significant per-cell MAE difference from EKFs given the rated capacity or tracking it online, and a lower inter-cell standard deviation of per-cell MAE for the deployed reservoir; capacity-aware EKFs remain more accurate on the reference cells. A causal online estimator and on-chip capacity-aware baselines are the next steps. The NASA PCoE dataset is public [17]; the design and procedure are given in Sections IV and V.

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